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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CS3214-1- NOSQL DATABASE

(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Combining sharding and replication provides:**
A) Only availability
B) Only consistency
C) Both scalability and fault tolerance
D) No benefit
2. **A single server model is limited because:**
A) It does not support ACID properties
B) It cannot scale beyond hardware limits
C) It uses only SQL
D) It enforces replication always
3. **Session consistency ensures:**
A) The same user sees a consistent view during a session
B) Data is consistent across all nodes instantly
C) Data is partitioned into sessions
D) Transactions are always serializable
4. **Redis is primarily a:**
A) Document database
B) Key-value store
C) Graph database
D) Column-family database
5. **Which of the following is NOT a challenge of MapReduce?**
A) High-latency batch processing
B) Difficulty in iterative computations
C) Poor handling of structured queries
D) Strong support for transactions
6. **The “Reduce” step in MapReduce is responsible for:**
A) Aggregating intermediate results
B) Splitting input data
C) Distributing tasks
D) Sharding data
7. **HBase is an example of:**
A) AP system
B) ACID system
C) CA system
D) CP system
8. **A system that prioritizes Availability and Partition tolerance may sacrifice:**
A) Latency
B) Consistency
C) Partition tolerance
D) Availability
9. **MongoDB stores data in the form of:**
A) Tables
B) Rows and columns
C) CSV files
D) JSON-like documents
10. **MongoDB is best suited for:**
A) Fixed schema data
B) Unstructured and semi-structured data
C) Complex joins
D) OLTP only
11. **Each MongoDB document contains a unique identifier field called:**
A) _key
B) id
C) _id
D) doc_id

12. In MongoDB, a "field" is equivalent to what in RDBMS?
 A) Row B) Column
 C) Table D) Index
13. Which of the following line skips the first 5 documents in the bios collection and returns all remaining documents?
 A) db.bios.find().limit(5) B) db.bios.find().skip(1)
 C) db.bios.find().skip(5) D) db.bios.find().sort(5)
14. In MongoDB, documents are represented in a format called:
 A) BSON B) XML
 C) CSV D) YAML
15. Which method is used to insert multiple documents?
 A) insertMany() B) insertMultiple()
 C) insertAll() D) saveAll()
16. What does this command do?
db.products.find({price: {\$gt: 100}})
 A) Finds products with a price greater than 100 B) Finds products with a price less than 100
 C) Finds products with a price greater than 100 D) Finds products with a price equal to 100
17. Cassandra is best suited for:
 A) Small datasets B) Real-time analytics only
 C) High write throughput and scalability D) In-memory data only
18. The data type used for storing collection of key-value pairs is:
 A) list B) set
 C) map D) dict
19. Which command is used to add a new column to an existing table?
 A) UPDATE TABLE B) MODIFY COLUMN
 C) ADD COLUMN D) ALTER TABLE ADD
20. Which command imports data from a CSV file into a Cassandra table?
 A) LOAD DATA B) COPY FROM
 C) IMPORT FROM D) INSERT FROM

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are NoSQL databases? Explain the different types of NoSQL databases.	6	L*2	1	1
	b) What is CAP theorem? Explain the trade-off between Consistency and Availability with respect to CAP theorem.	6	L3	2	2
	c) Explain the process of partitioning in MapReduce. How does it improve processing efficiency?	4	L3	2	2
2.	a) Explain the concept of sharding in NoSQL databases. How does it help in managing large datasets?	6	L2	1	1
	b) Explain two stage MapReduce with example.	6	L3	2	2
	c) Write the difference between SQL and NoSQL.	4	L3	2	2
3.	a) Explain aggregate data models with example.	6	L2	2	1
	b) Why are Partition and combiners used? Explain with example.	6	L3	2	2
	c) What is Impedance Mismatch? Explain with example.	4	L2	1	2
Unit – II					
4.	a) List and describe the various data types supported by MongoDB.	6	L3	3	2
	b) Create "Students" collection and store basic information about the student such as student roll number, name, grade, and hobbies.				
	i. Find the number of documents in the Students collection wherein the Grade is VII.				
	ii. Sort the documents from the Students collection first on Grade in ascending order and then on Hobbies in descending order.				
	iii. Display the last 2 documents from the Students collection.	6	L3	4	2

c)	What are aggregate functions in MongoDB, and how do they differ from those in SQL? Explain.	4	L2	4	1
5. a)	How Creation of data is done using insert(), update() method in MongoDB? Explain with query examples.	6	L2	3	2
b)	Consider the collection "Customers" with four customer documents that contains fields namely ,CustID, AccBal , and AccType. Write MongoDB query for the following cases				
i.	Group on "CustID" and determine the maximum "AccBal" for each group.				
ii.	Filter out the documents with "AccType:S" and then group it on "CustID" and compute the sum of "AccBal".				
iii.	Filter out the documents with "AccType:S" and then group it on "CustID" and then compute the sum of "AccBal" and then filter those documents wherein the "TotAccBal" is greater than 1200.	6	L3	4	2
c)	How can you store binary data (such as images or files) in MongoDB? Explain.	4	L1	3	1
6. a)	Explain the significance of '\$set', '\$push', and '\$pull' operators in array update with suitable query examples.	6	L3	3	2
b)	Write the MongoDB query for the following				
i.	Create student collection and store basic information about the student such as student roll number, name, grade, and hobbies.				
ii.	Update hobbies to "Chess" of the document with id 1 from the Students collection.				
iii.	Remove the document with the student's name from the Student collection.	6	L1	4	1
c)	How replication works in MongoDB? Explain.	4	L2	3	1
Unit – III					
7. a)	Design a column family to support the following requirement				
i.	Store basic information about the student such as student roll number, name, date of joining, and LastExamPercent.				
ii.	Delete the column "LastExamPercent" from the student information table for the record where the RollNo=2.	6	L2	5	1
b)	How does Cassandra handle collections (such as lists, sets, and maps) in CQL? Explain with example.	6	L2	5	1
c)	What is Cassandra? Explain its features.	4	L1	5	1
8. a)	List and describe the various data types available in CQL.	8	L2	5	1
b)	Design a column family to support the following requirement				
i.	Store basic information about the student such as student roll number, name, date of birth, and LastExamPercent.				
ii.	Store the subject preferences of each student. There should be a minimum of two subject preferences maximum of four. The order of preferences is given by students should be preserved.				
iii.	Store the hobbies of each student. There should be a minimum of two and a maximum of four. The hobbies should be arranged in alphabetical order.				
iv.	View the record from student information where RollNo has a value 1 or 2 or 3.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
